12/11/2025

Reactor Network Transient Response and Tuning

Prepared for: Webb Industries
Presented by: Anthony Jordan
Prepared by: Isaac Chambers

Burdell, Inc.
COMPUTING DIVISION

Major Deliverables

- **Steady State Analysis** (mg/m^3):
  - C1 = 11.5094, C2 = 11.5094, C3 = 19.0566, C4 = 16.9983, C5 = 11.5094

- **Transient Response Analysis** gives t90 (minutes):
  C1: t90 = 22.10, C2: t90 = 31.90, C3: t90 = 12.90, C4: t90 = 36.40, C5: t90 = 71.90

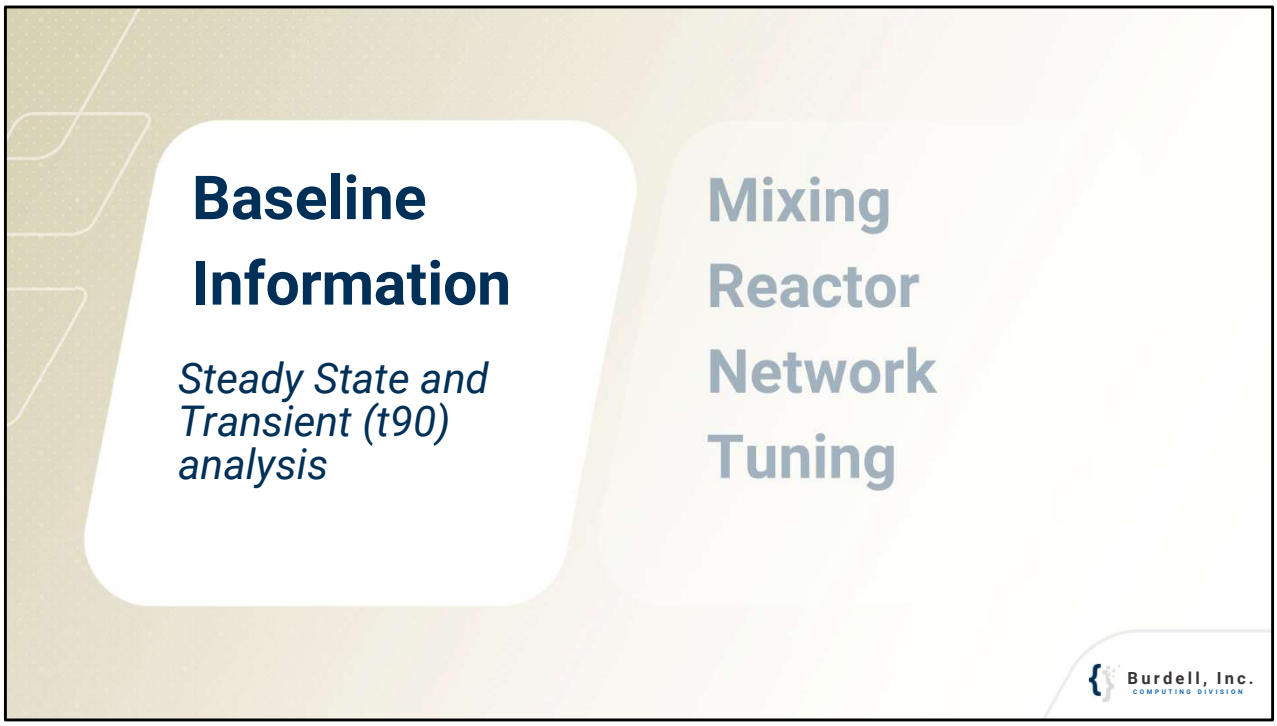
- **Mixing Reactor Network Tuning** gives flow rates (m^3/min): Q12 = 12.6562, Q15 = 10.2115, Q23 = 5.0625, Q24 = 0.2930, Q25 = 6.0513, Q31 = 17.0859, Q34 = 3.5000, Q44 = 13.7500, Q54 = 10.0854, Q55 = 6.7500 → **gives >10mg/m³ & < t90=30min**

Burdell, Inc.
COMPUTING DIVISION

For the steady-state analysis, the model shows that all five reactors settle between 11.5 and 19 mg/m3, which means the baseline system already meets the required minimum concentration of 10 mg/m3 required.

]For the transient response, several tanks rise too slowly, especially C4 and C5. C5 in particular has a t90 of 71.9 minutes, which is well beyond the 30-minute requirement. This confirms the dynamic performance is lacking, requiring the following tuning process.

The tuning process we implemented used a systematic eigenvalue based approach that will be discussed in detail later. The results give a set of volumetric flow rates shown, which satisfy steady state concentrations above 10mg/m3 and a t90 of less than 30 minutes to meet all specifications.



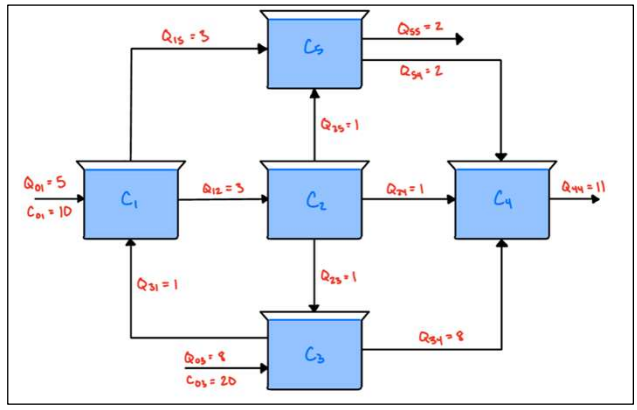
No presenter notes necessary for this slide unless there is something you want to say to introduce the topic.

Also note that this is not the only way to structure this slide – If you have more than two tasks or decide to use a different format to display them given an appropriate context, that is fine.

System and Assumptions

System Overview and Assumptions:

- 5 perfectly mixed continuous reactors
- Fixed volumes always due to implied dilutant flow
- Flows between them (inlets and exits)
- Concentrations measured in mg/m³
- Flows measured in m³/min
- Sensors accurate to 4 sig figs
- Fluids incompressible
- Perfect mixing inside each reactor
- Constant volume reactors
- All internal valves are controllable except Q01 and Q03
- Initial concentration in all reactors: 0 mg/m³
- No disturbances or time-varying flows



Burdell, Inc.
COMPUTING DIVISION

The system consists of five perfectly mixed reactors connected through a fixed network of internal flows, along with two external inlet streams and two outlet streams. Each reactor has a known, constant volume, and all flow rates are expressed in m3/min. Concentrations throughout the analysis are in mg/m3, and sensors in the real system are accurate to four significant figures. All reactors begin at zero concentration, which defines the transient startup condition.

To model this system, we make several assumptions that are both provided by the client and common engineering models. We treat the fluid as incompressible, we assume perfect mixing within each reactor so that the concentration is uniform at all times, and we assume no chemical reactions are occurring, making this a purely mixing and transport problem. The reactor volumes remain constant, meaning any excess inflow is implicitly balanced by a diluent outflow not shown in the schematic. Only the internal valves are adjustable during the tuning process.

These assumptions allow us to simplify our mass and concentration flows to be expressed as matrix equations that can be solved for the steady state analysis, transient simulation, and manipulated for later tuning. Next we will discuss the mathematical framework that we analyze.

Methods

Must determine steady state concentrations:

- Compute a linear mass balance for a matrix equation
  - Efficiently solved simultaneously in MATLAB with built in operation
- This mathematical formulation is valid due to the previously discussed assumptions.

$$V_i \frac{dC_i}{dt} = \sum_{\text{in}} Q_{\text{in}} C_{\text{in}} - \sum_{\text{out}} Q_{\text{out}} C_i$$

$$\frac{dC_i}{dt} = 0 \quad \Longrightarrow \quad 0 = \sum_{\text{in}} Q_{\text{in}} C_{\text{in}} - \sum_{\text{out}} Q_{\text{out}} C_i$$

$$\sum_{\text{out}} Q_{\text{out}} C_i = \sum_{\text{in}} Q_{\text{in}} C_{\text{in}}$$

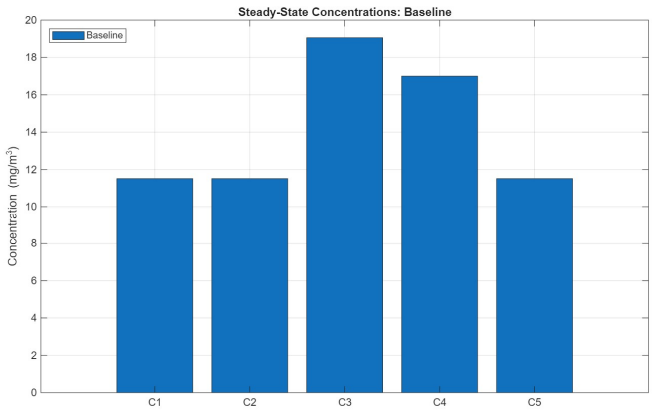
$$A\,\mathbf{C}_{ss} = \mathbf{b}$$

$$\mathbf{C}_{ss} = A^{-1}\mathbf{b}$$

Burdell, Inc.
COMPUTING DIVISION

We start from a standard mass balance on each perfectly mixed reactor. For tank iii, the rate of accumulation is equal to the total mass flow in minus the total mass flow out. Mathematically, that's [read shown on slide].
For the steady-state analysis, the client is explicitly interested in the conditions after the system has settled, so we set all time derivatives to zero. That reduces each mass balance to a simple algebraic relation: inflow of mass equals outflow of mass for each reactor.
When we write all five reactor equations together, we can express them compactly in matrix form as [read shown on slide], where Css is the vector of steady-state concentrations in each tank. The matrix A encodes the flow structure of the network, and b contains the external unmanaged inlet terms.
We then solve this system in MATLAB using the backslash operator, which is a numerically stable approach. This quickly computes the steady state concentrations.

Results



Steady State concentrations (mg/m^3):
C1 = 11.5094
C2 = 11.5094
C3 = 19.0566
C4 = 16.9983
C5 = 11.5094

Burdell, Inc.
COMPUTING DIVISION

The steady state analysis gives us the long-term concentrations in each reactor once all transients have died out. These values show that every reactor maintains at least 10 mg/m3 in steady operation.
From the results shown here, all five reactors meet that requirement: C1, C2, and C5 settle around 11.5 mg/m3, C4 is just under 17 mg/m3, and C3 is the highest at about 19 mg/m3. This confirms that the baseline network, even before any tuning, satisfies the steady-state specification, and that the majority of tuning will involve keeping these satisfied while focusing on the transient analysis discussed next.
It's worth noting that this behavior aligns with the structure of the flow network. Reactors with higher combined inflow g naturally settle at higher concentrations, and of course those with balanced inflow and outflow level out closer to the external inlet concentrations. Because the system is linear and we assume perfect mixing and no reactions, these values represent general long term behaviors that would be expected, and the results are sensible evaluations that give us a reliable baseline to build our analysis on.

Methods and Assumptions

We must compute the transient analysis t90 (time for each reactor to reach 90% of steady state), where dCi/dt is NOT 0:

- Method choice: custom built 4th-order Runge-Kutta function
  - Accurate fixed-step integration
  - Produces smooth trajectories for t90 calculation
  - The resulting C(t) is compared to 0.9*C_ss to determine t90
- Required assumptions
  - Same as listed for the System overview
  - Time invariant (constant flow rates)
  - No rapid function fluctuation

$$k_1 = f(t_n,\ C_n)$$

$$k_2 = f\left(t_n + \frac{h}{2},\ C_n + \frac{h}{2}k_1\right)$$

$$k_3 = f\left(t_n + \frac{h}{2},\ C_n + \frac{h}{2}k_2\right)$$

$$k_4 = f(t_n + h,\ C_n + hk_3)$$

$$C_{n+1} = C_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$A_{\text{dyn}} = -D^{-1}A, \qquad b_{\text{dyn}} = D^{-1}b$$

$$\frac{d\mathbf{C}}{dt} = A_{\text{dyn}}\,\mathbf{C} + b_{\text{dyn}}.$$

Burdell, Inc.
COMPUTING DIVISION

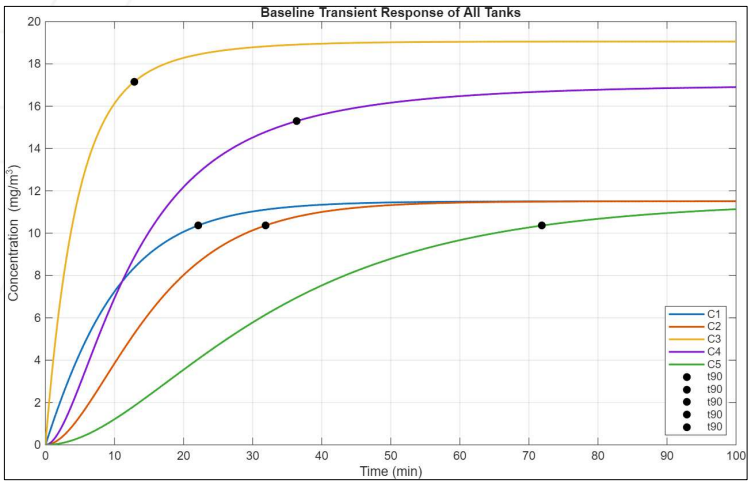
The reactor network can be written as a system of linear ODEs, except this time, the change in concentration as time change is not zero, and to simulate how concentrations evolve from startup we numerically integrate this system over time. Because the system is well behaved and there are no reactions, no stiffness, and all flows are constant, we use a fourth-order Runge Kutta method (as asked of us). RK4 gives a very accurate estimate of the solution with a fixed time step and avoids the overhead of MATLAB's adaptive solvers, which are unnecessary for a system this smooth (known from general knowledge of simple tank mixing systems).

At each step, RK4 evaluates four intermediate slopes, which gives a weighted average slope that approximates the true solution with fourth-order accuracy. This is a computationally inexpensive task as it is done with mainly with matrix operations. We apply this update repeatedly over the 0 to 100 minute window to obtain concentrations in all five reactors at each time point.

Once we have the time histories, computing t90 is done for each reactor by determining the earliest time at which the concentration reaches 90% of its steady-state value.

The baseline results show that several reactors, especially C4 and C5, rise too slowly. This is the main motivation for tuning, which we will examine next.

Results



Transient response, t90 (mins):
C1: t90 = 22.10
C2: t90 = 31.90
C3: t90 = 12.90
C4: t90 = 36.40
C5: t90 = 71.90

Burdell, Inc.
COMPUTING DIVISION

These results show how each reactor behaves dynamically when the system starts from zero concentration. The transient curves on the figure illustrate how quickly each tank approaches its steady-state value, and the listed t90 times quantify that performance. Note that the tanks with higher t90 times have a 'slower' curve, which is associated with a less negative eigenvalue, discussed next.

Some reactors rise quickly, but others, C4 and especially C5 are significantly too slow. C1 and C3 reach 90% of steady state within about 12–22 minutes, which is well below the client's 30-minute requirement. C2 is borderline at about 32 minutes, but C4 takes roughly 36 minutes, and C5 requires over 71 minutes to reach 90% of its steady-state concentration.

The slowest two tanks define the system bottleneck, and they also dominate the long-term behavior of the entire network. This result is the baseline for the following tuning work.

Baseline Information

Steady State and Transient (t90) analysis

Mixing Reactor Network Tuning

Burdell, Inc.
COMPUTING DIVISION

No presenter notes necessary for this slide unless there is something you want to say to introduce the topic.

Tuning and Eigenvalues

- After computing eigenvalues, the eigenvalue closest to zero gives the slowest eigenmode, and therefore the valves surroduning that tank / relating to it are the key focus

- In order to find the desired gradient direction to tune the system, a cost function was used to encourage movement in the right direction
- Workflow:
Scale one valve up or down
 Recompute C_ss, eigenvalues, and cost
 Accept change if JJJ decreases
 Halve step size and repeat until convergence
 Upstream valves adjust automatically to maintain Css≥10C_{ss} \ge 10Css≥10

$$J_{cost} = \alpha\, t_{90,\mathrm{eig}} + \beta \sum_i \max(0,\, 10 - C_{ss,i}) + \gamma \sum_v \left| \frac{Q_v - Q_v^{\mathrm{nom}}}{Q_v^{\mathrm{nom}}} \right|$$

Eigenvalues (descreasing order)
lambda_1 = -0.21177 +0.00000i
lambda_2 = -0.17746 +0.00000i
lambda_3 = -0.13750 +0.00000i
lambda_4 = -0.10578 +0.00000i
lambda_5 = -0.04000 +0.00000i

Slowest Eigenmode
Slowest eigenvalue (real part) = -0.040000
t90 from this mode = 71.9
Tank C1: mode amplitude = 0.000
Tank C2: mode amplitude = 0.000
Tank C3: mode amplitude = 0.000
Tank C4: mode amplitude = 0.256
Tank C5: mode amplitude = 1.000

Tanks to focus on:
(|mode amplitude| >= 0.25)
C4: |v| = 0.256
C5: |v| = 1.000

Burdell, Inc.
COMPUTING DIVISION

We combine all performance requirements into a single scalar cost. One part of the cost penalizes slow system dynamics using an eigenvalue-based rise-time estimate. Another part enforces the requirement that all steady-state concentrations remain at or above 10 mg per cubic meter. A third part prevents the optimizer from drifting toward unrealistic or extreme valve settings by penalizing deviations from nominal scales. Because this cost function includes the real physical constraints and engineering considerations, minimizing it naturally produces configurations that meet the client's goals without ad-hoc tuning.
Rather than adjusting every valve blindly, we identify which valves matter most by examining the eigenvector associated with the slowest dynamic mode. Large components of this eigenvector indicate where the slow dynamics physically occur in the network. In this system, tanks C4 and C5 dominate the slow mode. Mapping those tanks back to the flow network shows that Q55 and Q54 strongly influence turnover in C5, Q44 and Q34 affect C4, and Q15 and Q25 supply mass to those reactors. These valves therefore provide the greatest leverage for improving rise time. For the client, this means the approach is efficient and scalable, since only valves associated with large entries in the slow eigenvector ever need to be tuned.

The optimization itself proceeds using a coordinate-descent approach. For each selected valve, the flow is slightly scaled upward or downward. The dynamic matrices are then recomputed, the steady-state equations are solved to obtain the steady-state concentrations, and the eigenvalues are evaluated to update the cost function. If the cost decreases, the change is kept. If not, the change is reverted and the opposite direction is tested. After all selected valves are examined, the step size is reduced and the process repeats. This continues until the step size becomes very small and no further valve adjustment meaningfully reduces the cost, yielding the final tuned flow configuration.

Steady-state safety is automatically preserved through the cost function. If a valve adjustment causes any tank concentration to drop below 10 mg per cubic meter, the penalty term becomes large, making that adjustment unfavorable. This naturally forces the optimizer to increase upstream supply flows when needed, avoid over-draining tanks, and maintain feasible and safe steady-state concentrations. As a result, the client does not need to manually enforce constraints; the mathematics enforces them inherently.

The client can apply this method easily and repeatedly. As long as they can construct or update the system matrices, compute eigenvalues and eigenvectors, implement the cost function, and run the coordinate-descent loop, the framework remains valid. The method is modular, working for any reactor network size, transparent because every decision follows from measurable quantities, scalable from a handful of reactors to hundreds, and defensible because each tuning step is mathematically justified.

Tuning and Eigenvalues

Performance requirements: C_ss > 10mg/m^3, t90 < 30min

C(t) to the right describes the eigenvalue decomposition, meaning that:

$$C(t) = C_{ss} + \sum_{k=1}^{5} \alpha_k e^{\lambda_k t} v_k$$

- The lowest eigenvalue controls long-term rise time
- Its eigenvector identifies which tanks dominate the slow mode
- Tuning valves shifts more negative making a faster system

Burdell, Inc.
COMPUTING DIVISION

The transient behavior of a linear, perfectly mixed reactor network is governed by a first-order linear system. The time rate of change of the concentration vector is equal to a dynamics matrix multiplied by the concentration vector, plus a constant input term. This describes how concentrations evolve toward steady state.

This system admits a closed-form modal solution. The concentration at any time can be written as the steady-state concentration plus a sum of decaying modes. Each exponential–eigenvector pair represents one dynamic mode of the system. Because all eigenvalues are negative, the system is stable and every mode decays over time. The rate of decay of each mode is set by the real part of its eigenvalue.

The eigenvalue with real part closest to zero decays the slowest. This slow mode dominates the long-term behavior of the system. Even though faster modes die out quickly, the slow mode controls:

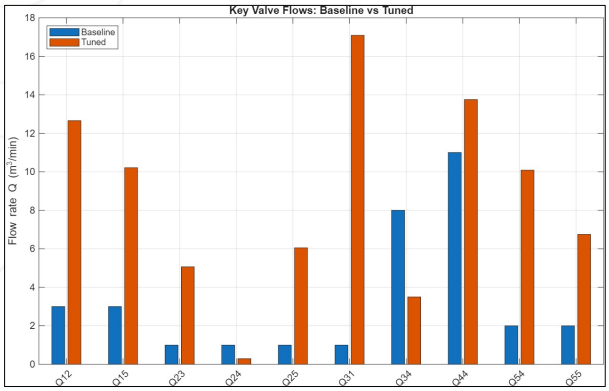
How fast concentrations approach steady state, how long it takes the system to reach 90 percent of steady state, and which tanks limit the performance of the entire network.

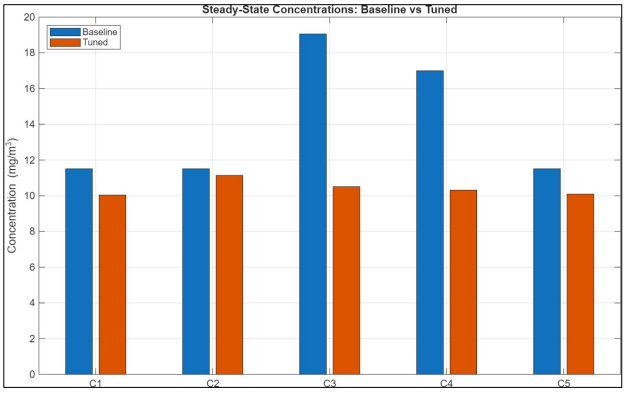
The eigenvector associated with the slow eigenvalue tells us where the slow dynamics physically occur in the system. If a particular tank has a large

component in this eigenvector, that tank is a bottleneck for the network.
In our case, the slow eigenvector has negligible components in tanks 1 through 3, a moderate component in tank 4, and a dominant component in tank 5. This means tanks 4 and 5 control the slow dynamics. As a result, flows associated with these tanks are the most effective tuning targets.

Increasing turnover in the problematic tanks increases outflow terms on the diagonal of the dynamics matrix and strengthens coupling between tanks through mass exchange. These changes push the slow eigenvalue to be more negative.
A more negative slow eigenvalue means faster exponential decay of the slow mode. This directly reduces the time it takes for the system to settle.

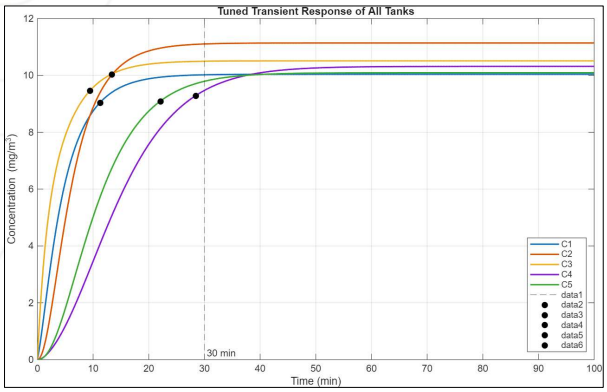
Results

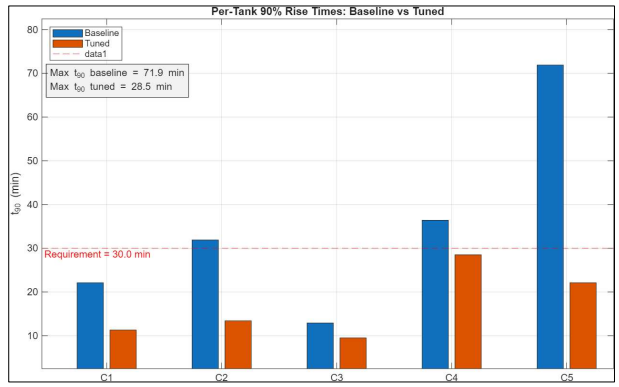




Burdell, Inc.
COMPUTING DIVISION

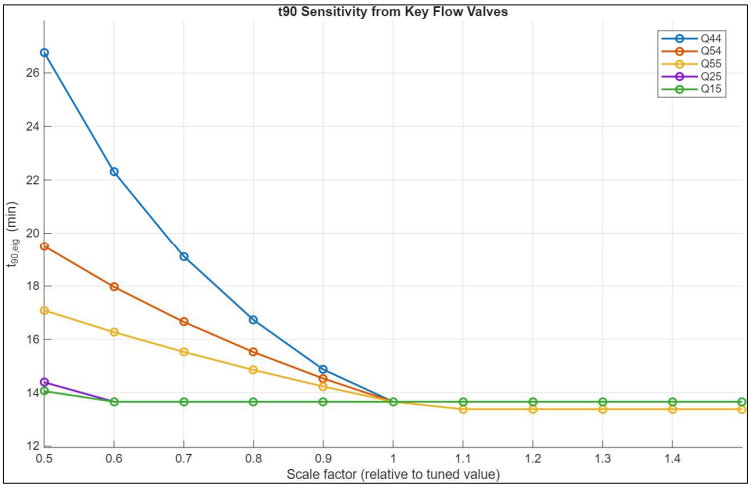
Results





Burdell, Inc.
COMPUTING DIVISION

Results



Burdell, Inc.
COMPUTING DIVISION

- **Steady State Analysis** gives a set of baseline concentrations

- **Transient Response Analysis** used a requested R.K.4 method to determine the baseline t90 values

- **Mixing Reactor Network Tuning** gives flow rates (m^3/min): Q12 = 12.6562, Q15 = 10.2115, Q23 = 5.0625, Q24 = 0.2930, Q25 = 6.0513, Q31 = 17.0859, Q34 = 3.5000, Q44 = 13.7500, Q54 = 10.0854, Q55 = 6.7500 → **Gives: >10mg/m^3 & < t90=30min**

Key Takeaway: The eigenvalue tuning process can be used for any similar problem!

Burdell, Inc.
COMPUTING DIVISION